

Fast Feature Field (F^3): A Predictive Representation of Events

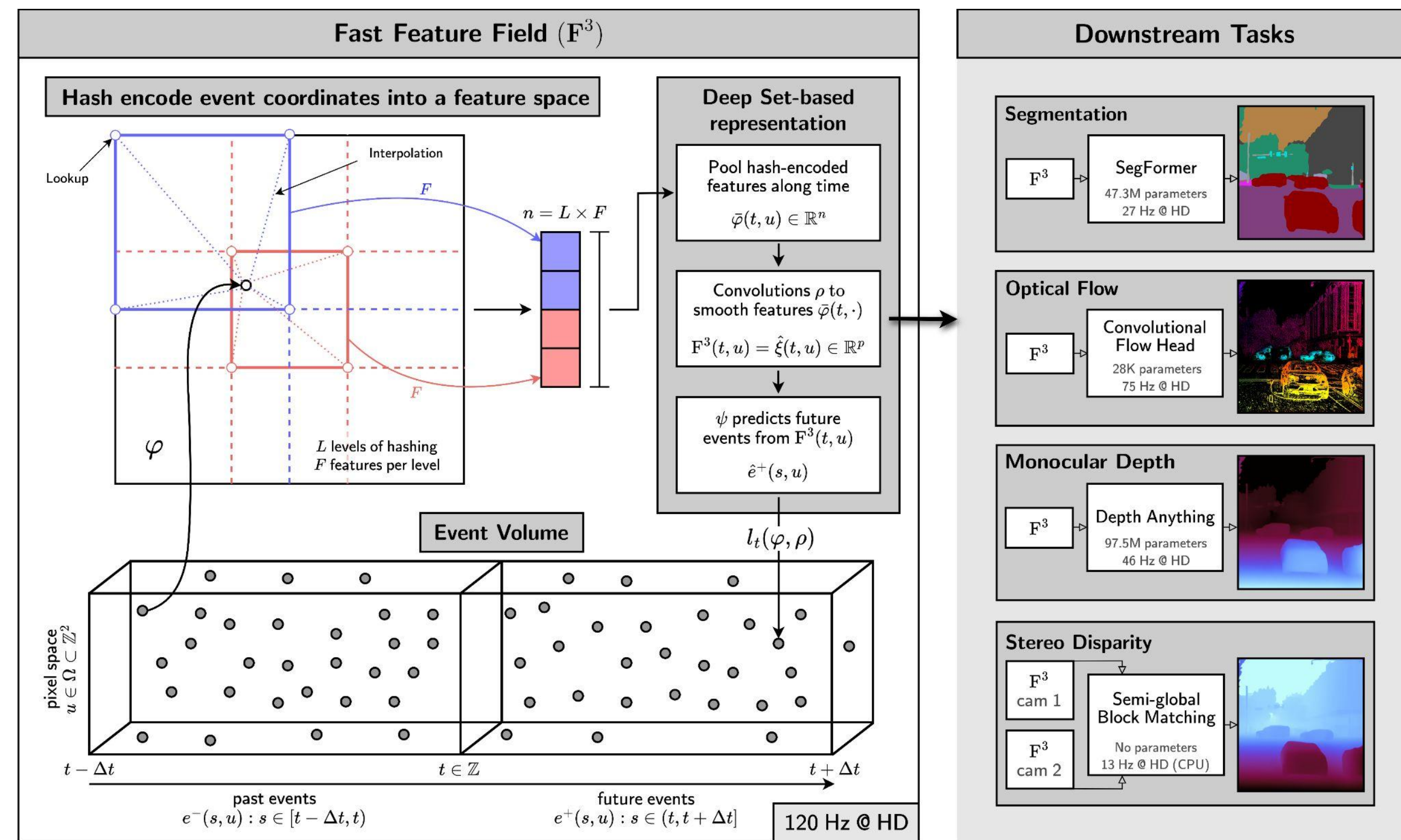
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Predicting future events is optimal for representation learning

★ Signals should be measured in a basis in which they are sparse.

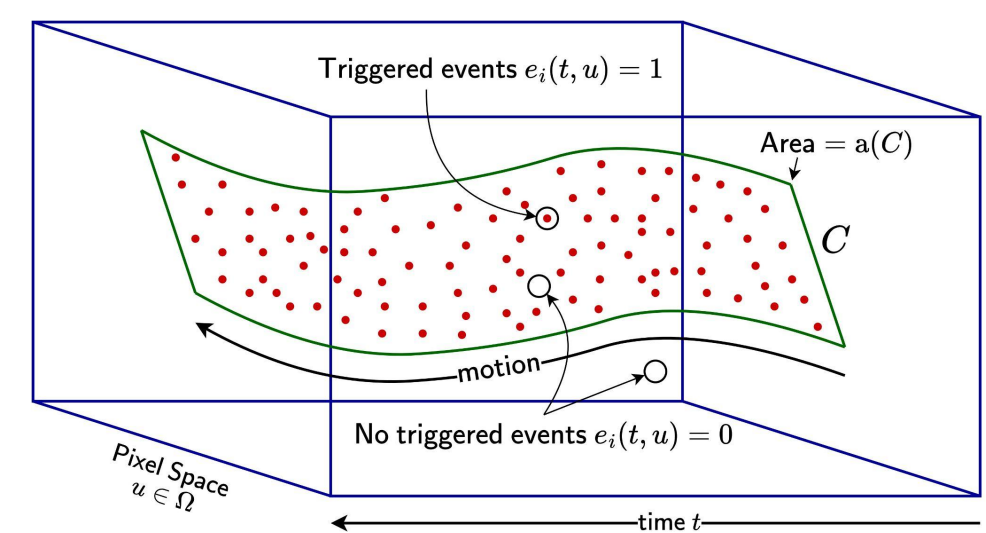
★ Basis functions predictive of future events implicitly encode structure and motion (evolution dynamics) of the signal. **Theorem stated below.**

F^3 instantiates the mathematical argument above to develop representations for events



Past events $e^-(s, u)$ between times $s \in [t - \Delta t, t]$ and pixels $u \in \Omega$ are encoded using a hash encoder, pooled across time, and smoothed in space to obtain $F^3(t, u)$ at a pixel u and time t . F^3 is efficient – the hash encoder only processes sparse events. F^3 is trained to predict future events $e^+(s, u)$ using a linear layer ($\hat{\psi}$). F^3 represents events as multi-channel image-like features – enabling many downstream tasks.

F^3 uses a dynamically weighted focal loss to build robustness to noise and event rates



Events (in red) on a spatiotemporal surface C . Theorem 2 in the paper states that if the cross-entropy loss class-weight (α) is too small, then the representation minimizing the objective predicts region C to have no events. Parameter α is chosen carefully to avoid such a trivial representation.

Why this loss?

Events are spatially **sparse** and temporally **stochastic** – autoregressive objectives are difficult to minimize.

- **Focal loss** handles calibration under sparsity. Maximum-entropy regularization in focal loss prevents F^3 from overfitting to noisy event data.

- **Class weight (α)** is dynamically chosen to be the fraction of voxels with events. This handles learning under temporal stochasticity of events.

Ideal denoising and spatiotemporal prediction of events using a library of bases Let ξ be a sufficient statistic of events that we want to estimate (e.g., F^3). For simplicity, assume events e to be Gaussian corrupted versions of ξ . Let ξ^-, ξ^+ denote ξ restricted to (past) $[t - \Delta t, t]$ and (future) $[t, t + \Delta t]$ respectively. Suppose $\xi^+ = A\xi^- + \zeta$, that is, ξ^+ evolves from ξ^- under linear dynamics A (with certain boundedness assumptions) and an additive Gaussian noise ζ . Under these assumptions, simultaneously identifying a basis to denoise events e^- and estimating the dynamics of the sufficient statistic ξ is an optimal strategy to recover ξ^+ from e^- .

Formally, the dynamics and the denoising basis

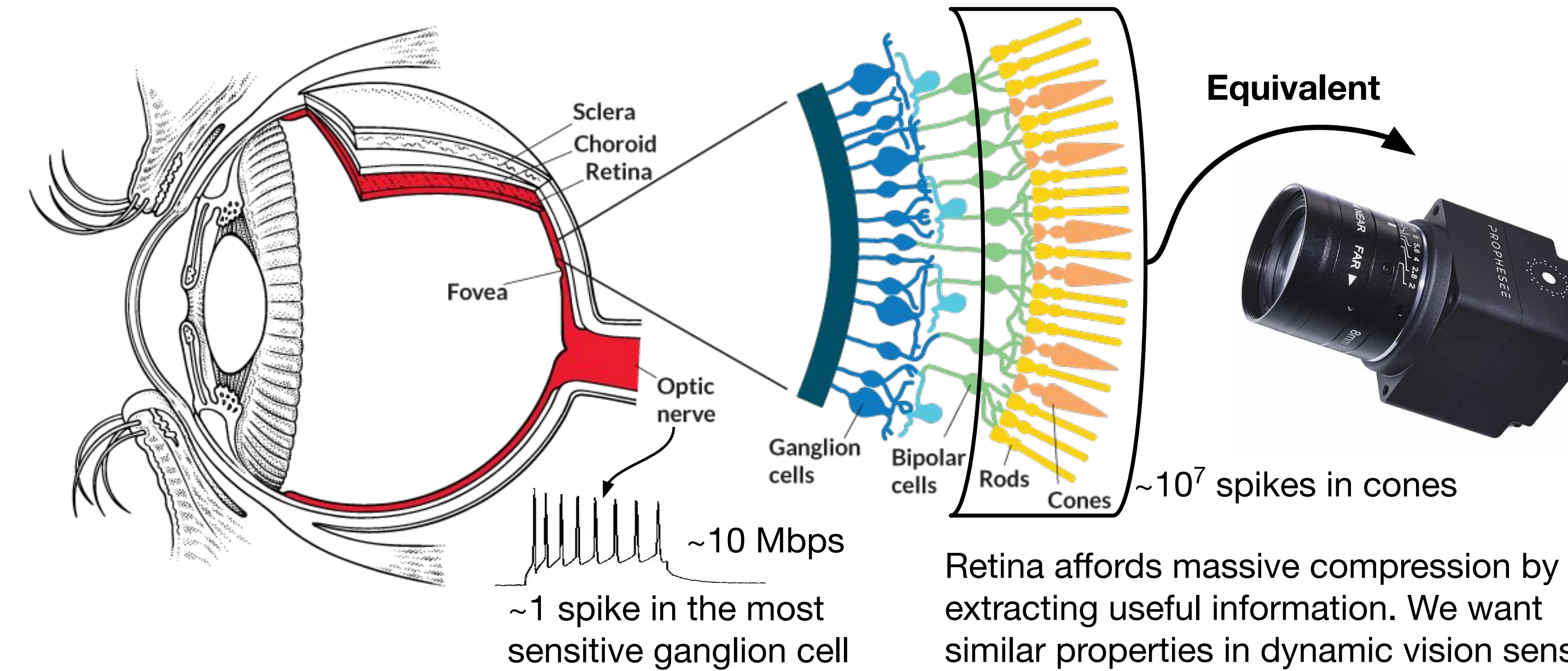
$$\hat{A}, \hat{B} = \argmin_{A, B \in \mathcal{L}} \left\{ \|e^+ - A\hat{\xi}^-\|_2^2 + \Lambda_n \|\hat{\xi}_B^-\|_0 \right\}$$

where $(\hat{\xi}_B^*)_i = \sigma_{\sqrt{\Lambda_n}}((e_B^*)_i)$ for $i \leq n$. $\mathcal{L}$ is a library of bases to search over. σ is a hard threshold non-linearity. Λ_n is a library $\mathcal{L}$ dependent constant, typically $\sim \log n$. $\hat{\xi}^-$ and $\hat{\xi}_B^-$ denote the same signal represented in the cardinal and B bases respectively. These quantities satisfy

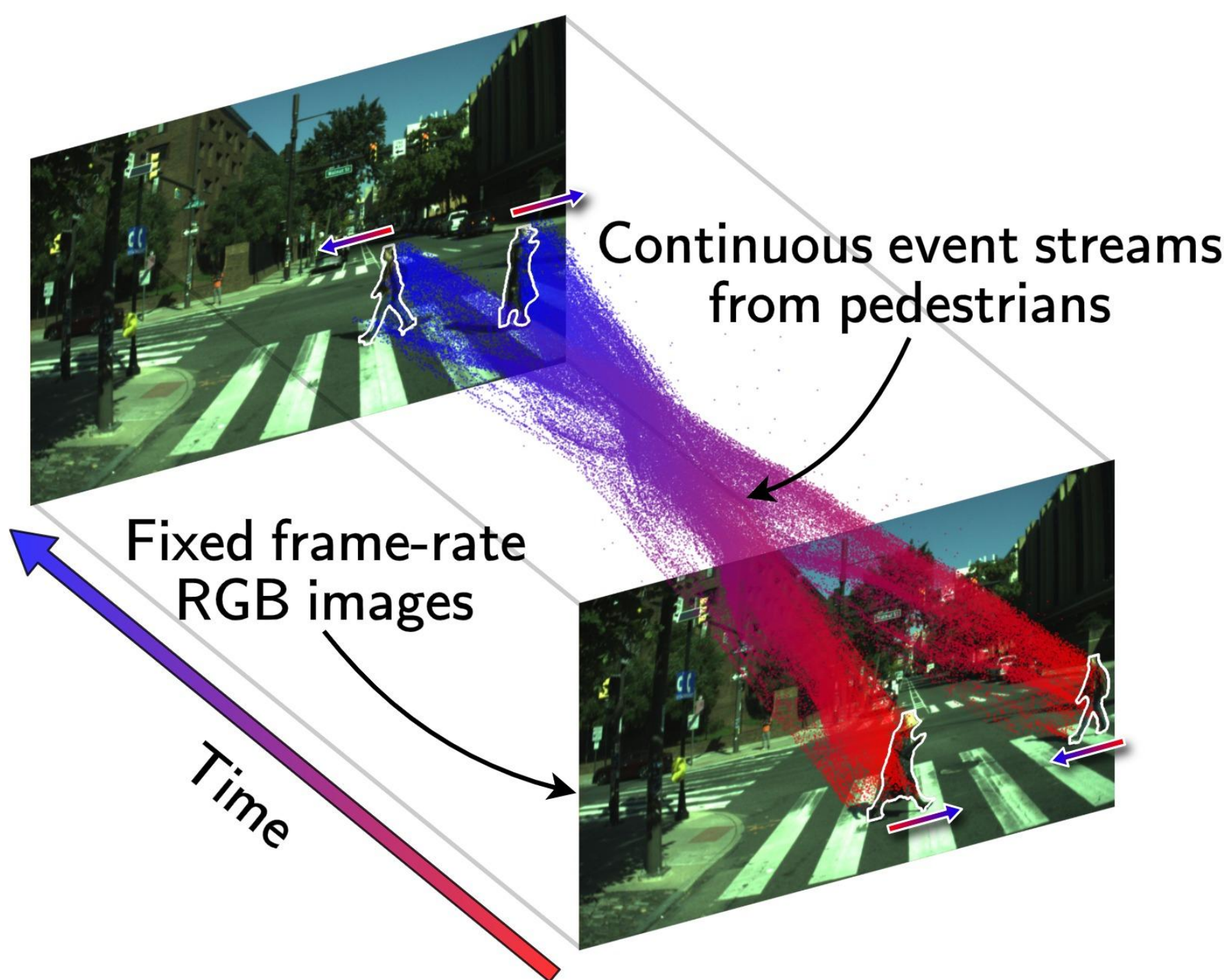
$$\|\xi^+ - A\hat{\xi}^-\|_2^2 \leq c_1 R_{\text{dynamics}}(\xi, \mathcal{L}) + c_2 \Lambda_n R_{\text{denoising}}(\xi^-, \mathcal{L}),$$

with probability at least $1 - ce/M_n$. Here c is a constant greater than 10 and M_n is the number of basis elements in $\mathcal{L}$, typically $\sim n \log n$. Constants c_1 and c_2 only depend on the assumed properties of the dynamics operator A . The quantities $R_{\text{dynamics}}(\xi, \mathcal{L})$ and $R_{\text{denoising}}(\xi^-, \mathcal{L})$ are oracle risks that depend upon (i) the ideal denoising basis for past events e^- , (ii) the coordinates i where $(e_B^-)_i$ is large, and (iii) the true future statistic ξ^+ .

Event cameras mimic the rods/cones of the mammalian retina

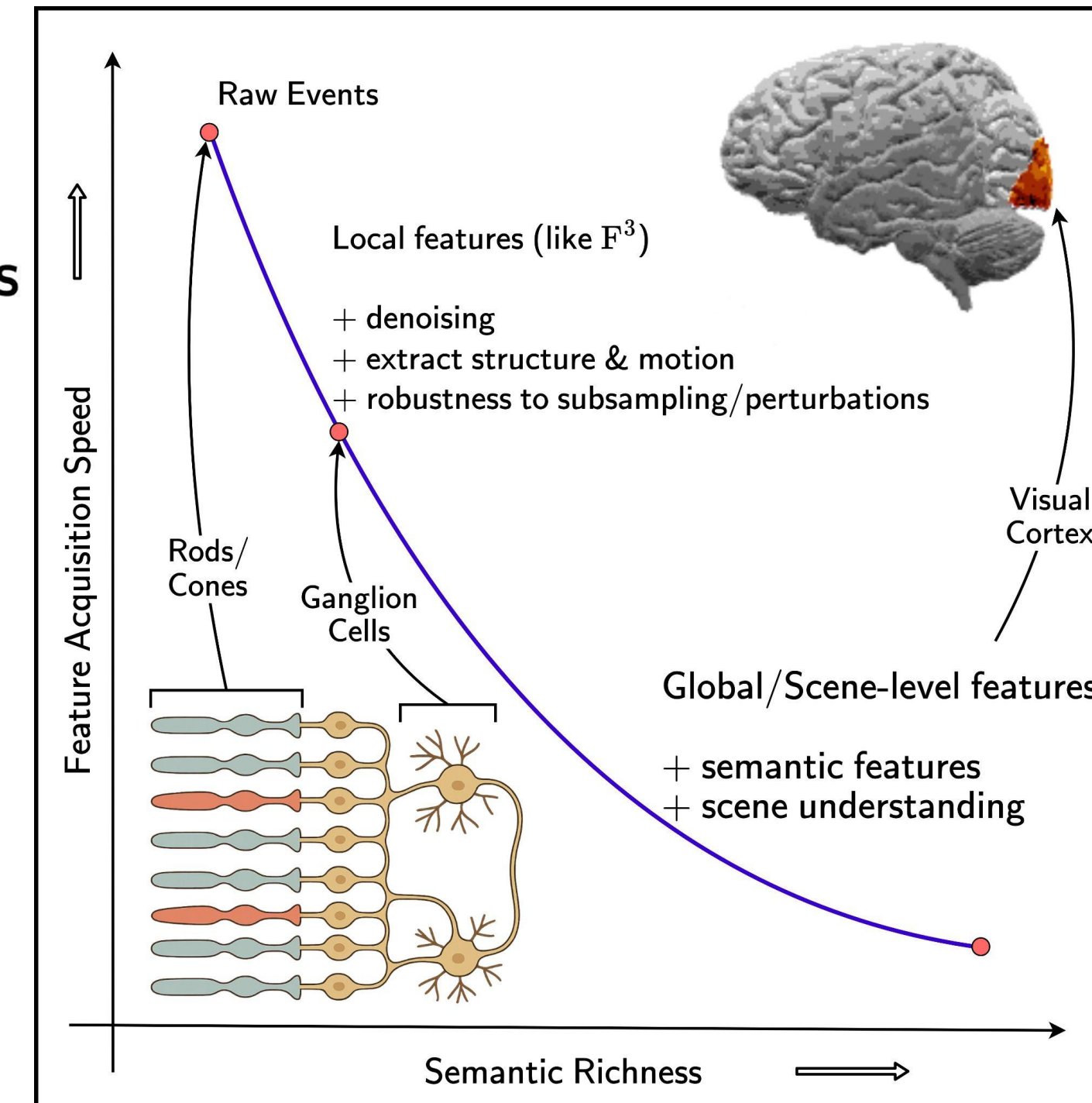


Events capture the scene without temporal aliasing



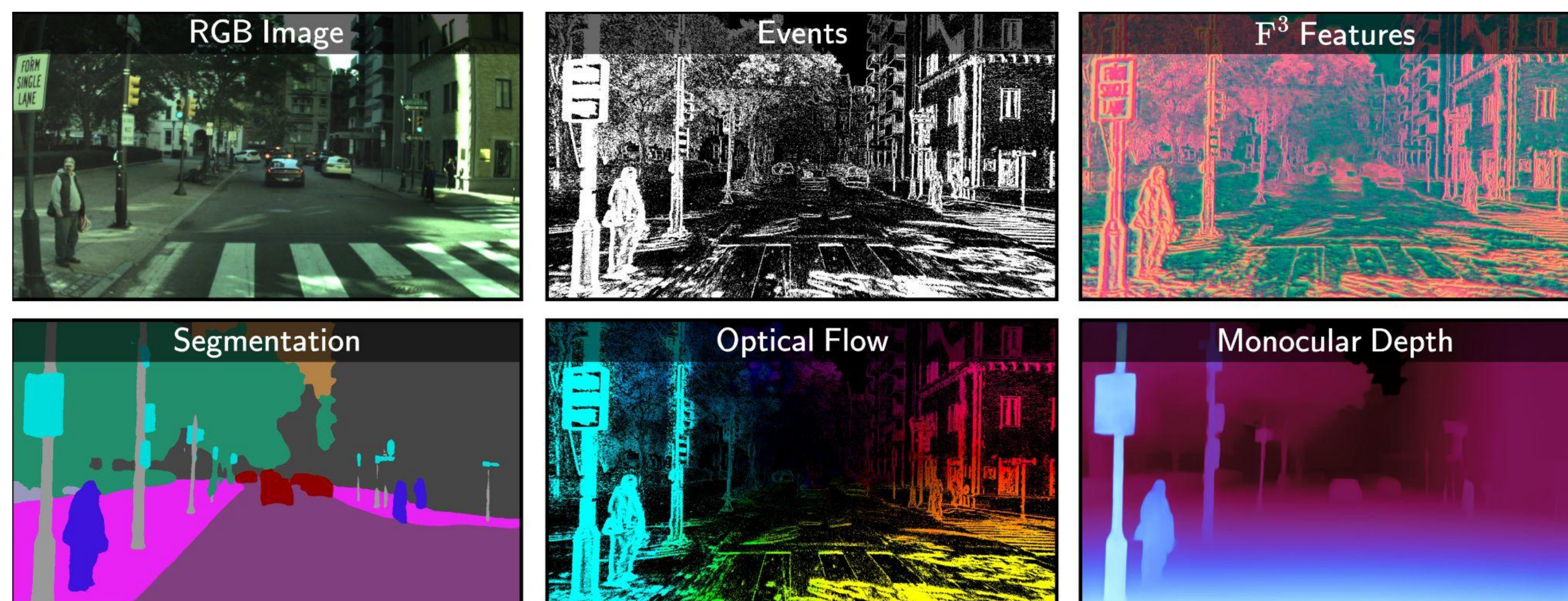
Unlike RGB cameras, event cameras are asynchronous sensors with high temporal resolution ($\sim 1 \mu s$), dynamic range (~ 120 dB) and a respectable spatial acuity (~ 1 MP).

F^3 brings events closer to retinal outputs



Beyond rods/cones, retinal neurons reduce redundancy and noise – eventually culminating in more semantic features in the visual cortex.

F^3 is a general representation – can be used in any standard computer vision algorithm or architecture



A typical urban driving scene with an RGB image, events from the past few milliseconds (displayed as an image), F^3 features (three principal components) that capture salient aspects of the scene such as edges, moving objects, and road signs, segmentation masks, optical flow (colors denote direction), and monocular depth estimate.

F^3 is fast and outperforms existing methods

How fast can an F^3 process 20 ms of events?

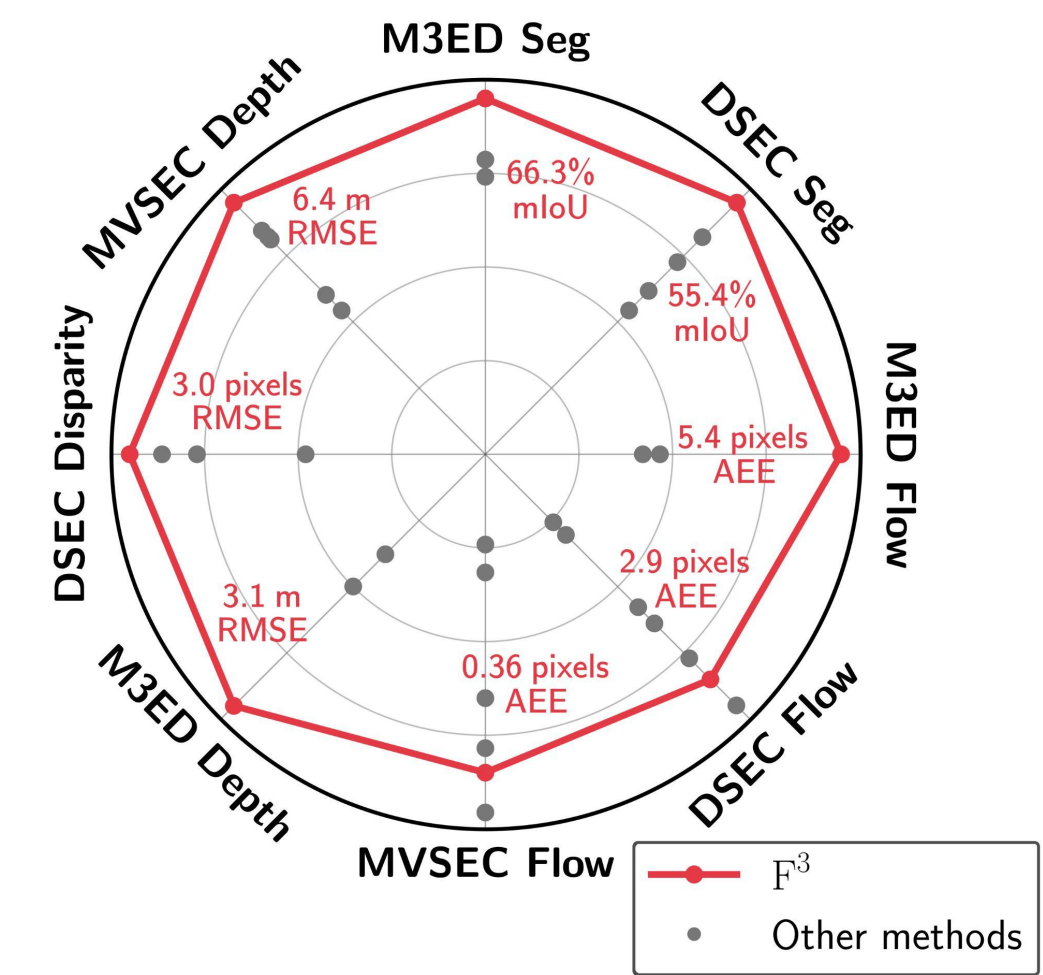
On an **RTX 4090**:

- 450 FPS @ HD (1280x720)
- 720 FPS @ VGA (640x480)

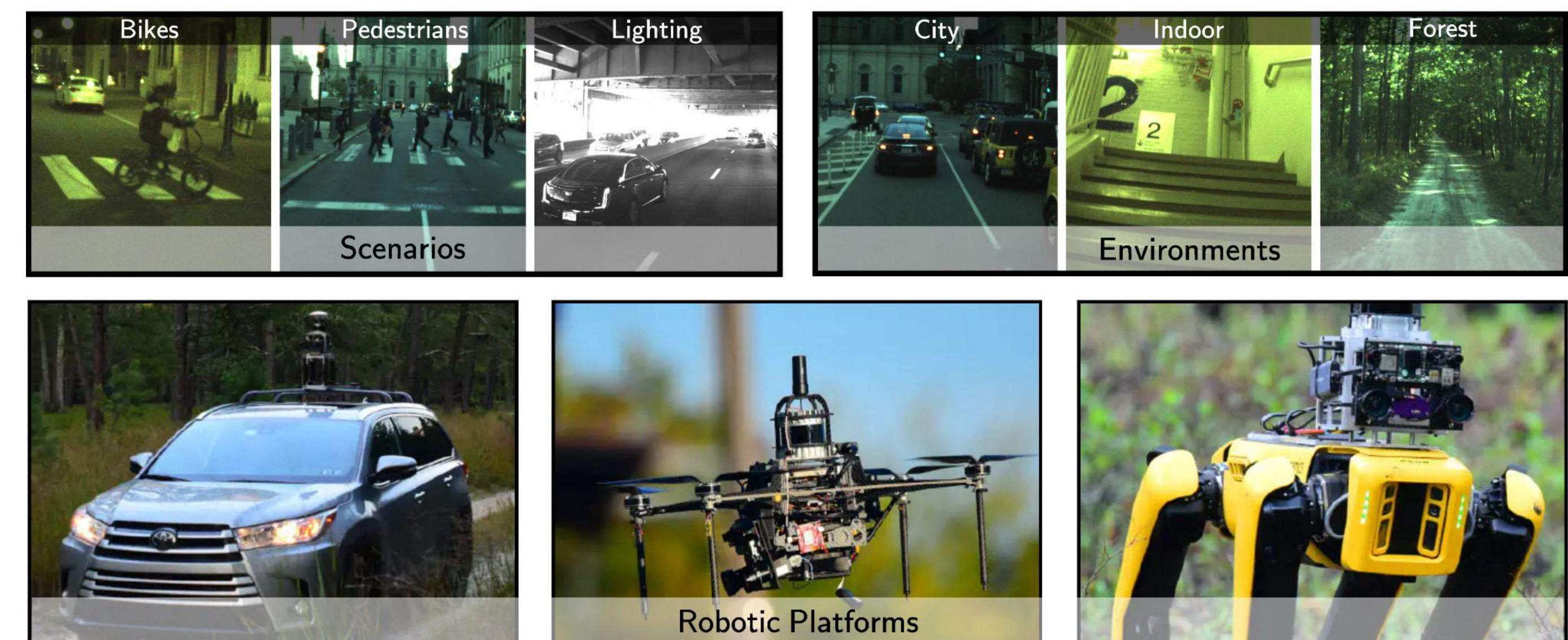
On a **Jetson Orin Nano**:

- 220 FPS @ (320x240)

F^3 outperforms existing methods (in gray) on downstream tasks like semantic segmentation, optical flow estimation and monocular depth estimation – while being **2-5x** faster.



F^3 generalizes to multiple environments, lighting and robotic platforms



We study F^3 -based representations in urban driving scenarios, forested environments, and indoor navigation, across different lighting conditions (high dynamic range, daytime, night-time), and across three different robotic platforms (a stereo event camera mounted atop a car, a flying platform and a quadruped robot).

Examples of F^3 -based predictions in a variety of tasks, scenes and scenarios

